Speed-Matching DCC Locomotives Using the STRR Speed Matching Table (SMT)

Rev2 10-17-2025-EWB

This STRR Speed Matching Table can do the following tasks with Locomotives (or rolling stock) having DCC decoders:

- **Provide a safe, dedicated loop of track to warm up, test, and diagnose locomotives and rolling stock without causing issues on the mainlines of the STRR layouts.**

- **Make changes to DCC decoders (Sound and non-sound) using a dedicated programming track and DecoderPro software.**

- **Speed Match or Measure Diesel and Steam DCC Locomotives to be able to run multiple units (MU) of different types of engines with reduced forces on the couplers.**

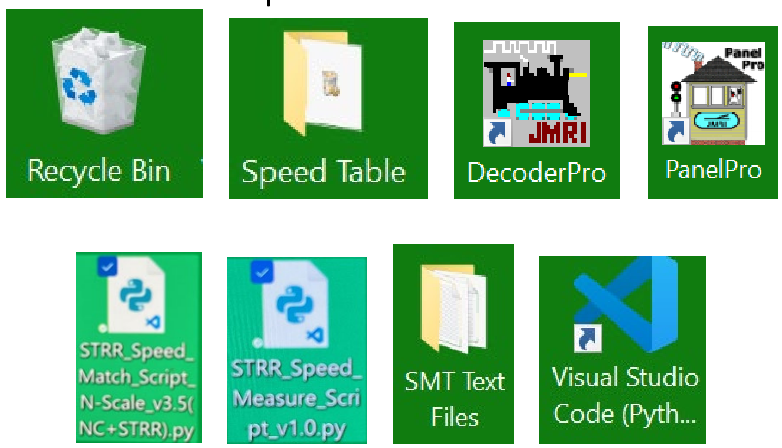
Section 1: Using the Speed Table Software to Speed Match Locos Automatically.

Section 2: Using the Speed Table Software to Speed Measure Locos Manually.

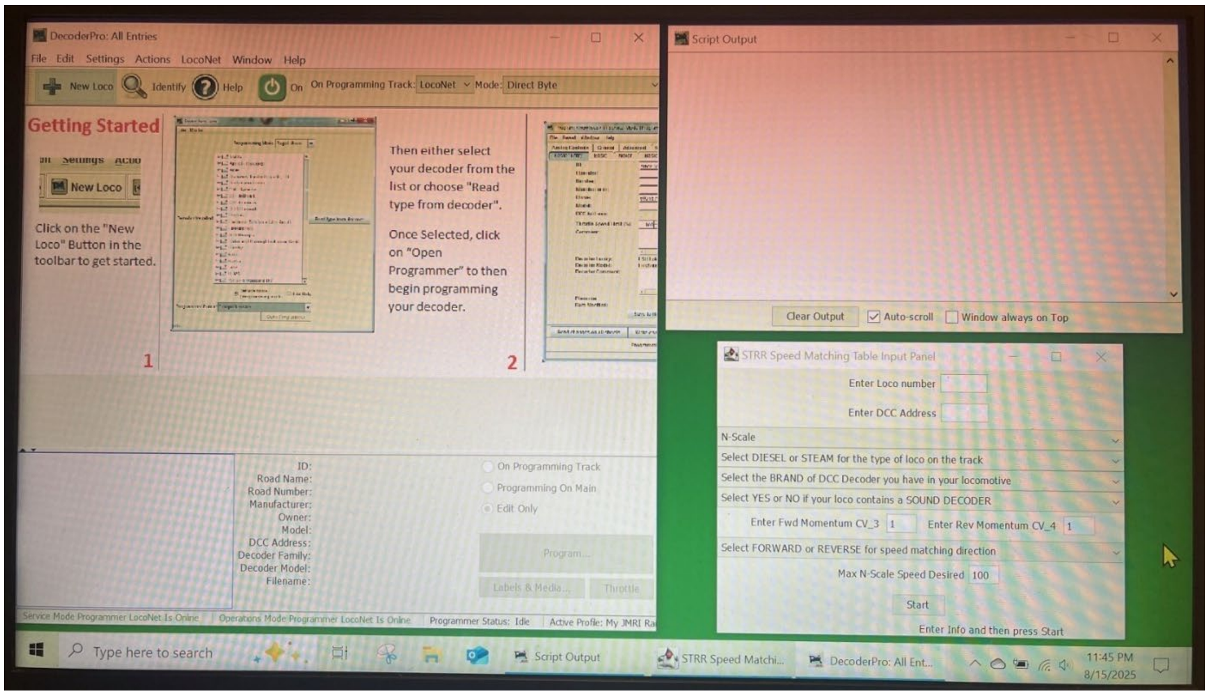
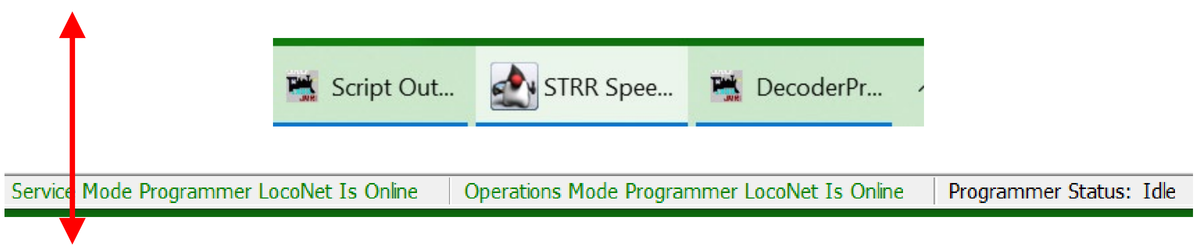
Section 3: Assembly of Speed Matching Table (SMT)

(Perform Section 3 in *REVERSE ORDER*, doing *THE REVERSE* of each Step to Disassemble the SMT.)

<u>Section 1:</u> Using the STRR Speed Matching Table Software to Automatically Speed MATCH DCC Locomotives.

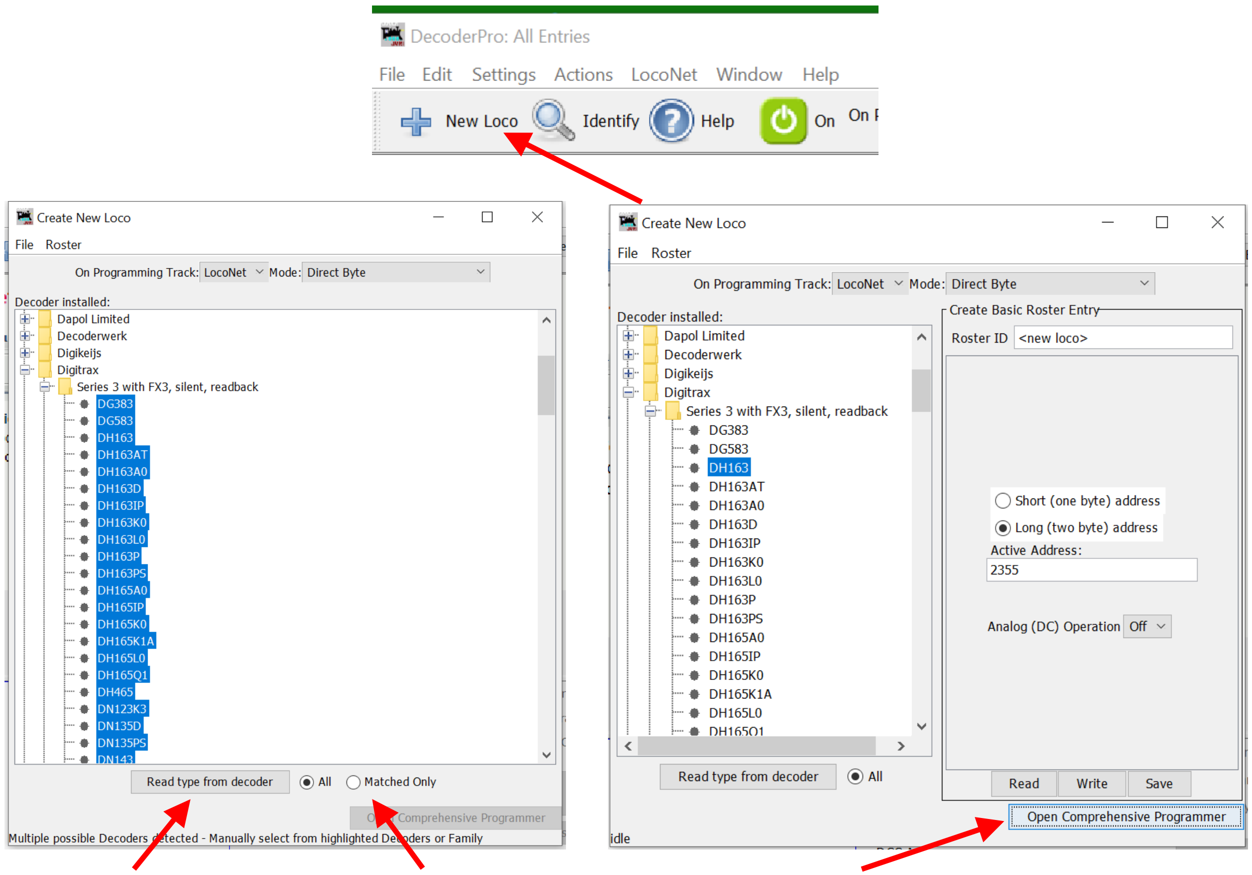
Steps	Programming Directions	Notes
Step 1	Familiarize yourself with icons on the desktop screen of the laptop. Below are a list of the key icons and their importance.  	You will want to make sure the Table Power Strip, Digitrax Command Station 100, BDL 168, PTB-100, LocoBuffer-USB, and Laptop are powered on. Look for green LED lights to be on and/or blinking. There will be 2 profiles options to select: 1 for <u>DecoderPro – Speed Matching</u> and 1 for <u>DecoderPro – Speed Measuring.</u>

<u>Recycle Bin</u> – Where files go to die.
<u>Speed Table Folder</u> – Stores original Speed Matching Script(s) and other key files for SMT.
<u>DecoderPro</u> – Used for SMT Speed Matching program and making changes to DCC decoders.
<u>PanelPro</u> – Used for making changes to DCC and layout panels and other key tasks associated with STRR layouts.
<u>STRR_Speed_Match_Script_N-Scale_v3.5(NC+STRR).py</u> – Python script used for Speed Matching DCC locomotives.
<u>STRR_Speed_Measure_Script_v1.0.py –</u> Python script use to manually measure loco speeds.
<u>SMT Text Files Folder</u> – Stores the text file from Speed Matching script that contains info on the loco programmed on SMT.
<u>Visual Studio Code</u> – Software used to edit python scripts.

Step 2	Put the DCC decoder-equipped locomotive on the programming track and **<u>double-click DecoderPro and select “Speed Matching”</u>** profile in the JMRI window that comes up. Three (3) windows will open up. You will know everything is working correctly because you will see **<u>3 task buttons in the taskbar</u>** and **<u>green text at the bottom of DecoderPro</u>** telling you everything is online. Do not close any of these!   	You can open these up separately clicking on the button in DecoderPro called “**<u>Script Output Window</u>**” to open the Script Output window. To run the SMT script, go to “**<u>Actions → Run Script</u>**” and browse to the STRR Speed <u>Matching</u> script stored on the desktop screen shown in Step 1.

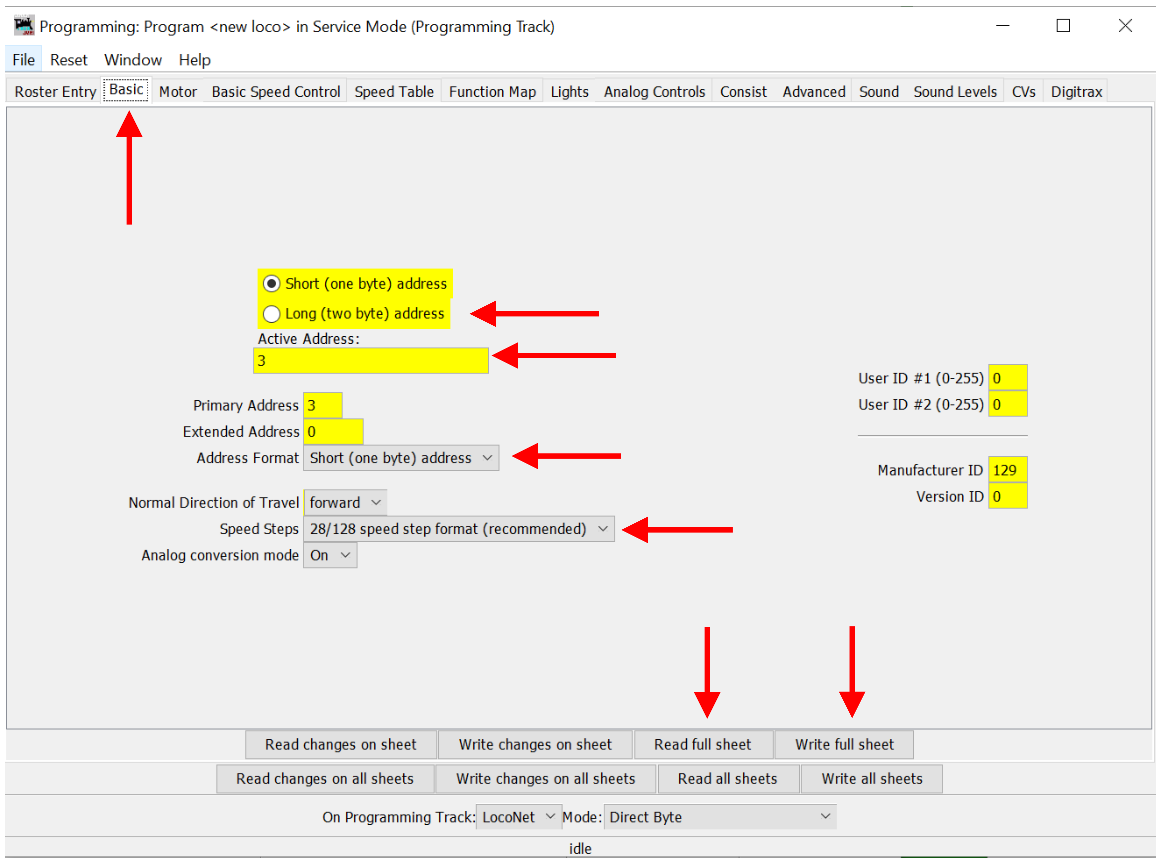
Step 3

In the **<u>DecoderPro</u>** window, Click on "**<u>New Loco</u>**" icon next to plus symbol (+). Click "**<u>Read type from decoder</u>**" icon to determine what DCC Decoder you have in the loco. Several options may be highlighted. If so, select the one that <u>best matches</u> your DCC decoder. Click on "**<u>Open Comprehensive Programmer</u>**".



If you select the radial dial that says "**<u>Matched Only</u>**", you will only see the options associated with the DCC chip manufacturer and DCC decoder series.

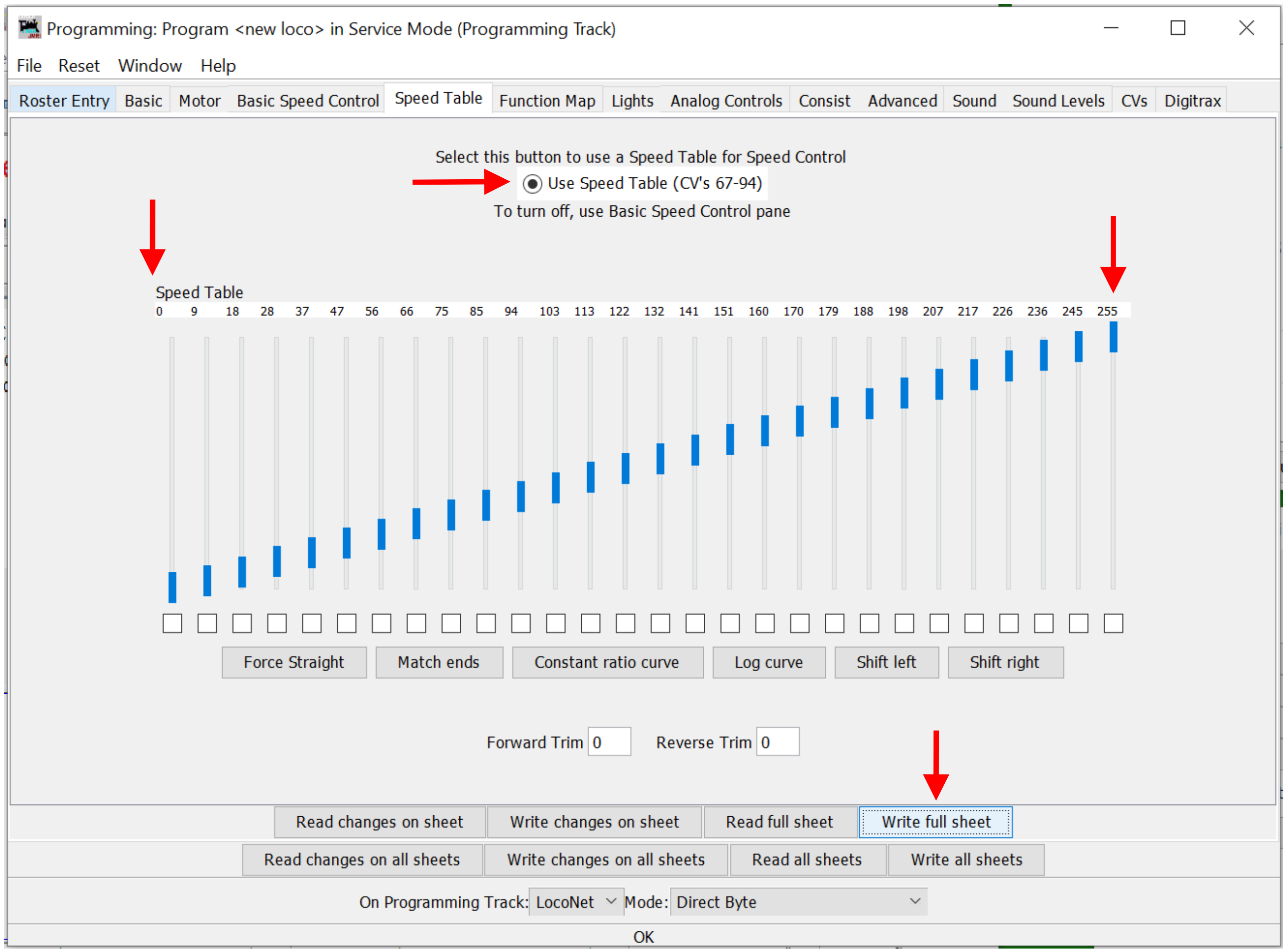
If you already know which type/model of DCC decoder you have, you can just click through the boxes to find the right one.

Step 4	Under the **"<u>Basic</u>"** tab, change/confirm the following values by clicking on "**<u>Read full sheet</u>**" at the bottom of the screen: → **Long (two byte) address:** <u>Checked</u> (this is for 4-digit DCC address) → **Active Address:** User-determined DCC address for loco (able to change!) → **Address Format:** Select **"**<u>Long (two byte) address"</u> → **Speed Steps:** Set to "<u>28/128 speed step format</u>" Click "**<u>Write full sheet</u>**" to record all values shown. 	These settings will enable the DCC loco to be programmed by SMT software. You can change these back to what you want after speed matching is finished. Depending on your DCC decoder, the "**<u>Basic</u>**" tab might have slightly different wording for the boxes shown.

Step 5

Under the "**Speed Table**" tab, you should see the following chart showing small slider bars going from bottom left at 0 to upper right at 255. Forward Trim and Reverse Trim should be set to 0. The radial dial for "**Use Speed Table (CV's 67-94)**" should be selected.

Click "**Write full sheet**" to record all values shown.



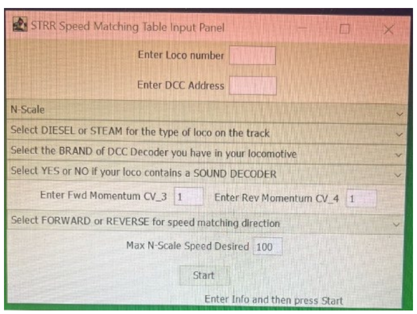
Step 6

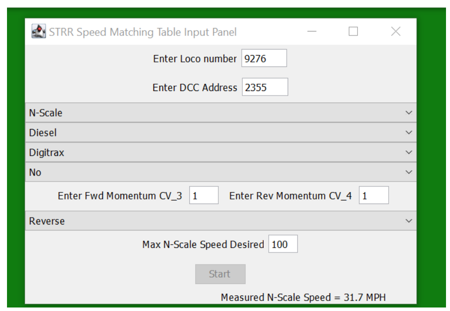
On the STRR Speed Matching Table Input Panel window in <u>bottom right of desktop</u>, please make a selection by clicking on the <u>drop-down arrows on the right **for each row**</u>.

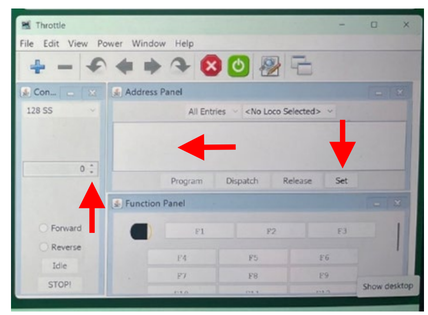
The input panel image on the <u>left</u> shows <u>the types of values to determine</u>. The input panel image on the <u>right</u> shows what a <u>completed input panel</u> looks like.

Choosing a direction to perform the speed matching (<u>Forward or Reverse</u>) is helpful if you are designating a loco to travel in a specific consist direction most of the time.

*****When speed matching different locos, it is best to keep maximum speed the same for all. This will allow you to mix and match locos regardless of decoder and model.*****







In the JMRI virtual throttle window, enter DCC number of loco and hit "<u>**Set**</u>" button. Confirm loco is called up correctly by moving the throttle slider (or clicking on toggle buttons) to confirm the loco moves correctly.

If you want your <u>Digitrax throttle</u> setting to correspond to locomotive speed, <u>set Max N-Scale Speed to **99 mph**</u>. *Throttle setting of 45% will equal approximately 45 n-scale mph for DCC loco.* <u>**This is the desired STRR club standard.**</u>

Selecting Diesel will allow the speed matching to occur in the <u>forward</u> **<u>OR</u>** <u>reverse</u> direction.

Selecting Steam will allow speed matching to occur <u>in the forward direction ONLY!</u>

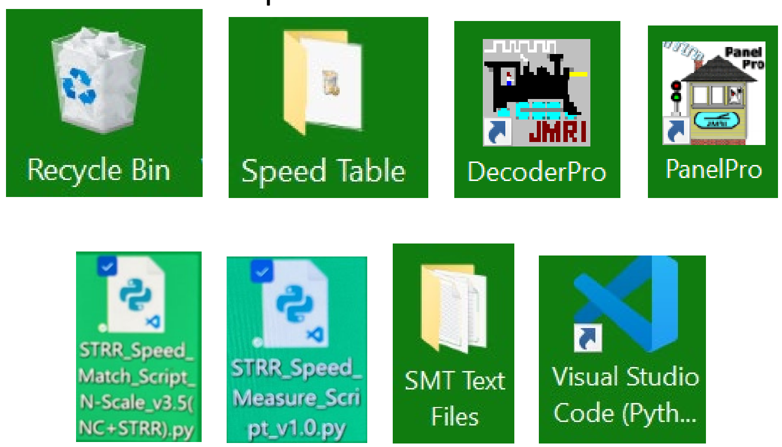
Step 7	When you click on "<u>**Start**</u>" on the STRR Input Panel, the following actions will take place: • Messages will begin to appear on the "<u>**Script Output**</u>" in the upper right of the desktop screen. • The loco will run a few times around in each direction to warm up. • The software will determine the maximum speed in each direction. • The software will determine specific speed steps using the block detection circuit to measure time and distance to get a calculated n-scale MPH reading. • The software will calculate the DecoderPro Speed Table values and program your locomotive DCC decoder with the appropriate CV value. • The program is finished when "<u>Test Time = X minutes</u>" is seen on the "<u>**Script Output**</u>" window. • A <u>text file</u> will be stored in the <u>folder on the desktop</u> called "**SMT Text Files**". This is for your records to keep! See Step 1 for reference. • Depending on the loco, DCC decoder, and max speeds determined, speed matching a loco could take between 12-25 minutes. (The lowest speed takes the longest to achieve!) • Clicking on the "<u>**Stop Sign**</u>" on JMRI throttle will cause the loco to stop. You will then need to close and reopen DecoderPro to start again.	<u>If you want to stop the software</u>, you need to click on the "Stop <u>**Sign**</u>" on JMRI throttle and then close and reopen DecoderPro to start again. To release the loco, hit the "Release: button on the virtual JMRI throttle.

Step 8	When speed matching a DCC locomotive **<u>with a SOUND Decoder</u>**, you will want to place your recently speed-matched locomotive on the programming track and **<u>go back to Steps 3 & 5 previously</u>** to verify that the DecoderPro Speed Table was recorded correctly. If not, you will need to manually enter the numbers from the saved text file and then select “**<u>Write the sheet</u>**” to the locomotive. **Non-Sound decoders usually do not have this issue.**	Make sure the radial dial for “Programming Track” is selected in DecoderPro.
Step 9	In DecoderPro, in the Comprehensive Program, under the tab “**<u>CVs</u>**”, one can see all the values recorded for each CV. By selecting “**<u>Read full sheet</u>**”, you will see each CV value recorded in the DCC Decoder. When **<u>CV29</u>** is set to “**<u>50</u>**”, it means the following: → 28/128 Speed Table Enabled → DC Analog <u>Disabled</u> ➔ Normal direction of travel is <u>Forward</u>. → <u>4-Digit</u> DCC Address being used. →*<u>Reverse direction</u>* sets CV 29 to **<u>51</u>**. ***<u>SEE STEP 10 FOR OTHER VALUES FOR CV29 AND THEIR EFFECTS.</u>***	

Step 10

CV Value for CV29	Speed Steps / Speed Table	Analog Mode Conversion	Normal Direction of Travel	2- or 4-Digit Address
006	28/128	ON	Forward	2
007	28/128	ON	Reverse	2
016	14 Speed Table	OFF	Forward	2
017	14 Speed Table	OFF	Reverse	2
018	28/128 Speed Table	OFF	Forward	2
019	28/128 Speed Table	OFF	Reverse	2
020	14 Speed Table	ON	Forward	2
021	14 Speed Table	ON	Reverse	2
022	28/128 Speed Table	ON	Forward	2
023	28/128 Speed Table	ON	Reverse	2
032	14	OFF	Forward	4
033	14	OFF	Reverse	4
034	28/128	OFF	Forward	4
035	28/128	OFF	Reverse	4
036	14	ON	Forward	4
037	14	ON	Reverse	4
038	28/128	ON	Forward	4
039	28/128	ON	Reverse	4
048	14 Speed Table	OFF	Forward	4
049	14 Speed Table	OFF	Reverse	4
050	28/128 Speed Table	OFF	Forward	4
051	28/128 Speed Table	OFF	Reverse	4
052	14 Speed Table	ON	Forward	4
053	14 Speed Table	ON	Reverse	4
054	28/128 Speed Table	ON	Forward	4
055	28/128 Speed Table	ON	Reverse	4

Section 2: Using the STRR SMT to Manually Speed MEASURE locomotive speeds.

Steps	Programming Directions	Notes
Step 1	Familiarize yourself with icons on the desktop screen of the laptop. Below are a list of the key icons and their importance.  	You will want to make sure the Table Power Strip, Digitrax Command Station 100, BDL 168, PTB-100, LocoBuffer-USB, and Laptop are powered on. Look for green LED lights to be on and/or blinking. There will be 2 profiles options to select: 1 for <u>DecoderPro – Speed Matching</u> and 1 for <u>DecoderPro – Speed Measuring.</u>

Recycle Bin – Where files go to die.
Speed Table Folder – Stores original Speed Matching Script(s) and other key files for SMT.
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SMT Text Files Folder – Stores the text file from Speed Matching script that contains info on the loco programmed on SMT.
Visual Studio Code – Software used to edit python scripts.

<table>
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Step 2

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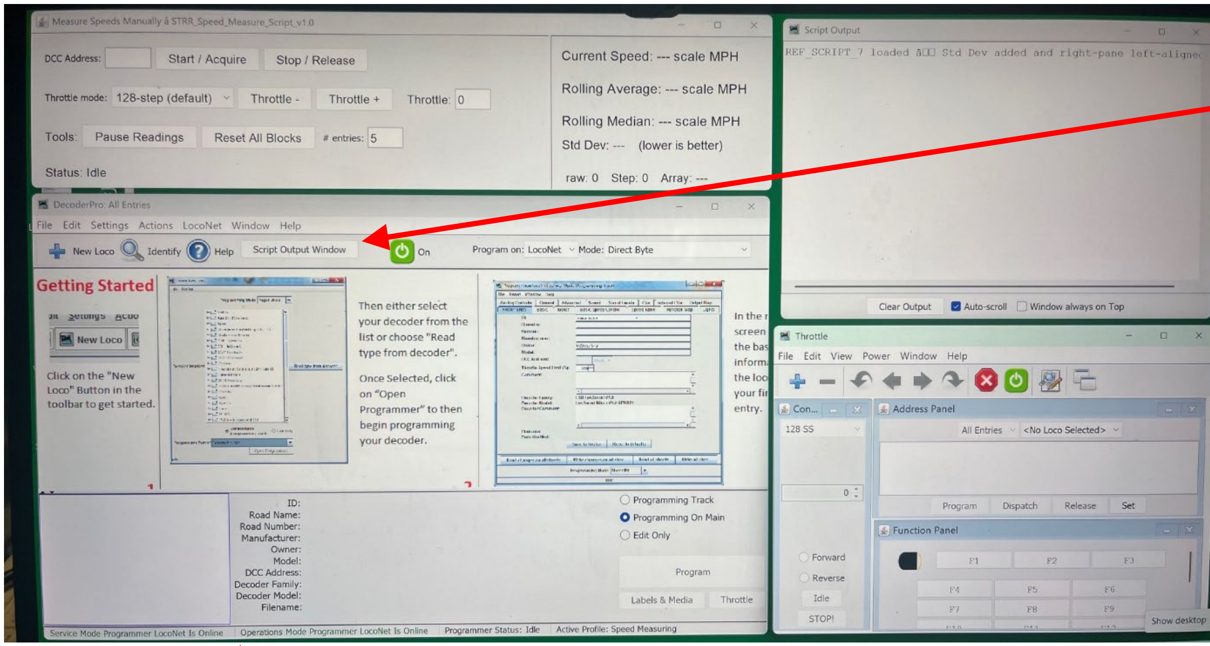
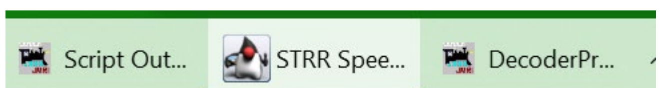
Put the DCC decoder-equipped locomotive on the programming track and **double-click DecoderPro and select "Speed Measuring"** profile in the JMRI window that comes up. Three (3) windows will open up. You will know everything is working correctly because you will see **3 task buttons in the taskbar** and **green text at the bottom of DecoderPro** telling you everything is online. Do not close any of these!





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You can open these up separately clicking on the button in DecoderPro called "**Script Output Window**" to open the Script Output window.

To run the SMT script, go to "**Actions → Run Script**" and browse to the STRR Speed Measuring script stored on the desktop screen shown in Step 1.

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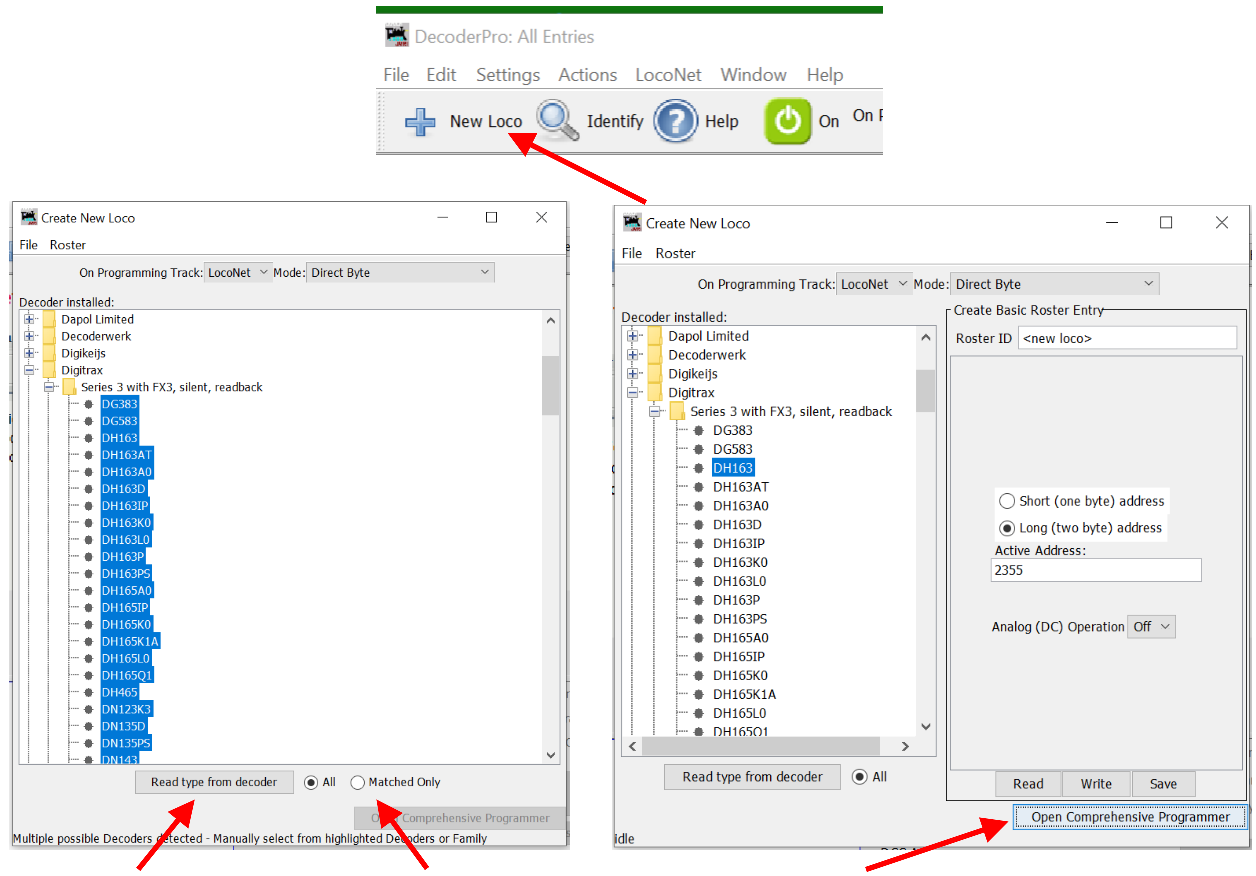
Step 3

Select the radial dial for "**<u>Programming Track</u>**". In the **<u>DecoderPro</u>** window, Click on "**<u>New Loco</u>**" icon next to plus symbol (+). Click "**<u>Read type from decoder</u>**" icon to determine what DCC Decoder you have in the loco. Several options may be highlighted. If so, select the one that <u>best matches</u> your DCC decoder. Click on "**<u>Open Comprehensive Programmer</u>**".



If you select the radial dial that says "**<u>Matched Only</u>**", you will only see the options associated with the DCC chip manufacturer and DCC decoder series.

If you already know which type/model of DCC decoder you have, you can just click through the boxes to find the right one.

Step 4

Under the **“<u>Basic</u>”** tab, change/confirm the following values by clicking on “**<u>Read full sheet</u>**” at the bottom of the screen:

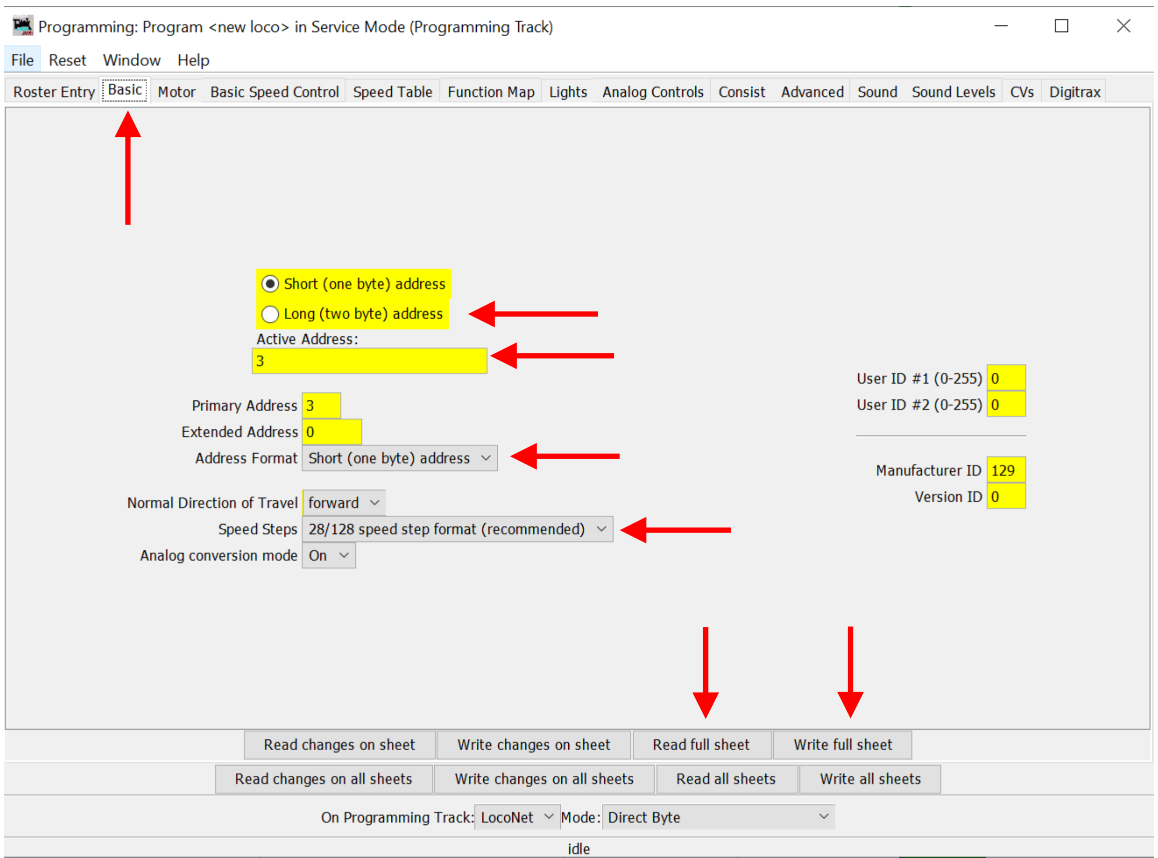
→ **Long (two byte) address:** <u>Checked</u> (this is for 4-digit DCC address)

→ **Active Address:** User-determined DCC address for loco (able to change!)

→ **Address Format:** Select **“**<u>Long (two byte) address”</u>

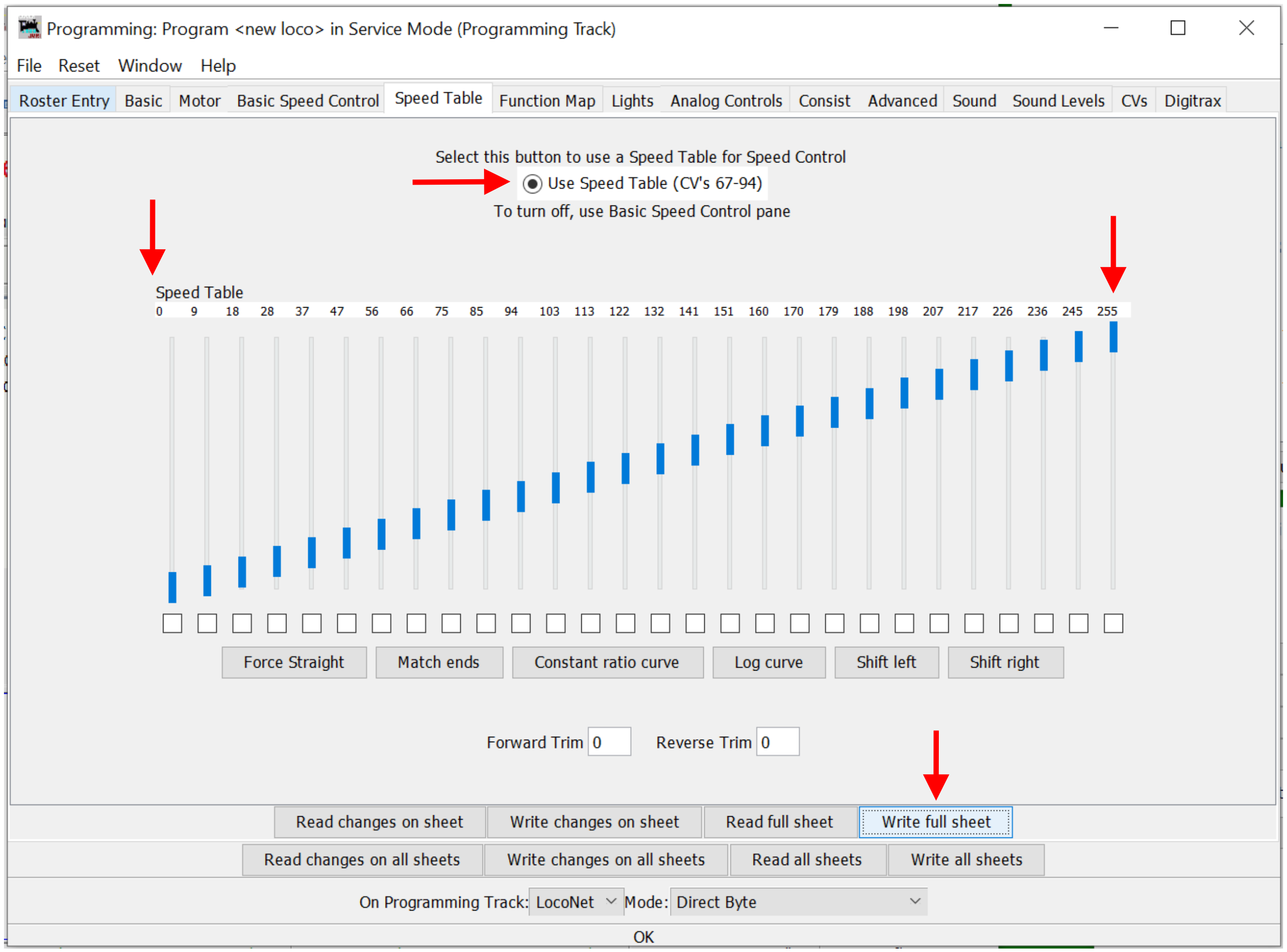
→ **Speed Steps:** Set to “<u>28/128 speed step format</u>”

Click “**<u>Write full sheet</u>**” to record all values shown.



These settings will enable the DCC loco to be programmed by SMT software. You can change these back to what you want after speed matching is finished.

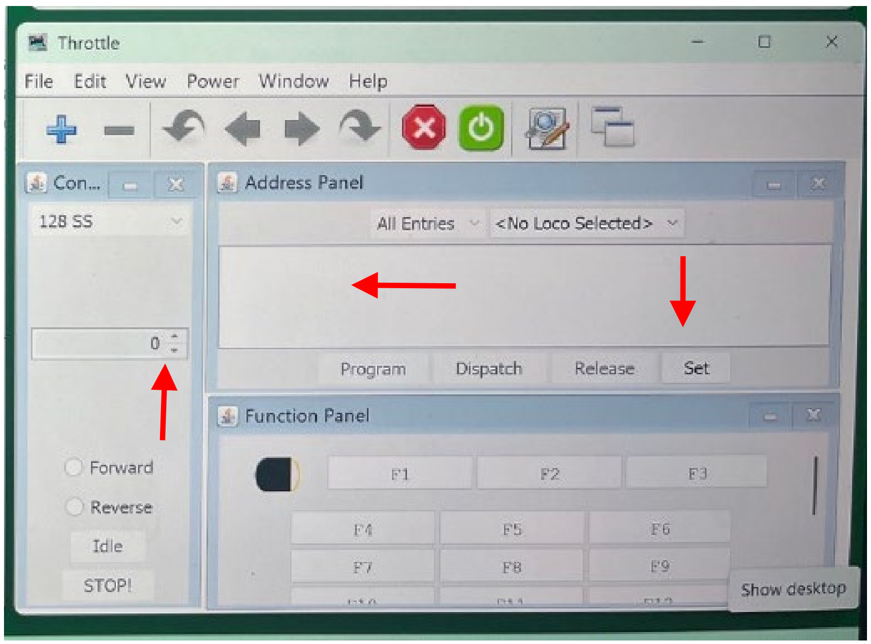
Depending on your DCC decoder, the “**<u>Basic</u>**” tab might have slightly different wording for the boxes shown.

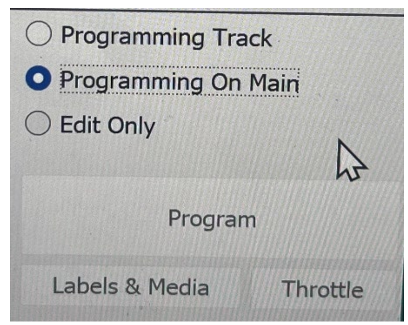
Step 5	Under the **"<u>Speed Table</u>"** tab, you should see the following chart showing small slider bars going from bottom left <u>at 0</u> to upper right <u>at 255</u>. Forward Trim and Reverse Trim should be <u>set to 0</u>. The radial dial for "**<u>Use Speed Table (CV's 67-94)</u>**" should be selected. Click "**<u>Write full sheet</u>**" to record all values shown. 	

Step 6

In the JMRI virtual throttle window, enter DCC number of loco and hit "**<u>Set</u>**" button. Put the loco on the loop of track so that <u>the direction you are measuring</u> **<u>is in the clockwise direction</u>**, select the direction button you wish to move, and move the throttle slider (or clicking on toggle buttons) to confirm the loco moves correctly.

Close out DecoderPro Programmer window and select the <u>radial dial</u> called "**<u>Programming on Main</u>**" (**<u>POM</u>**). Click "**Program**" Button to reopen <u>Programmer window</u>. Navigate to the **<u>Speed Table</u>** Tab.





If you have a roster shown in the throttle window, you can just select the loco you wish to connect to the throttle.

When programming a locomotive, it is important to understand what track you are using and which DecoderPro programmer you are using. POM allows you to write to a DCC decoder, but not read from it when a loco is on the loop of track.

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Step 7

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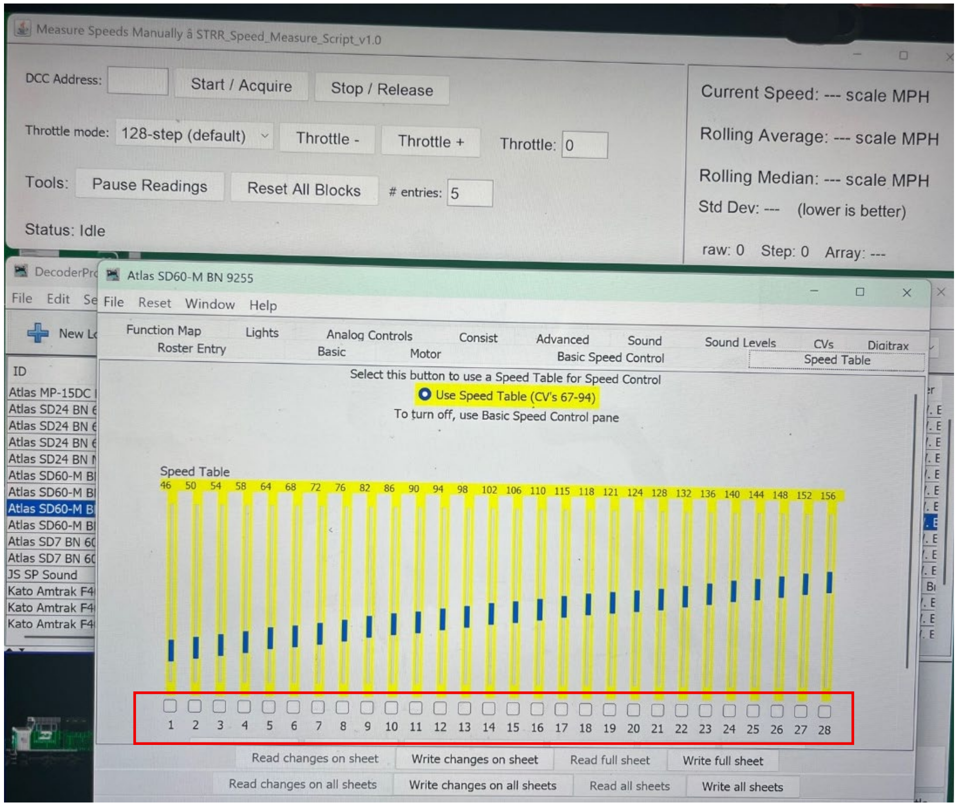
Arrange the Programmer and Measuring Speed windows on the screen so they look like the picture below. Notice the 1-28 speed steps below the speed table bar chart. See red box.



As you increase the speed of the JMRI virtual throttle, you will see the loco move in a clockwise direction and then speeds will begin to be shown in the script speed measuring panel.

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<u>If you want to stop the software</u>, you need to click on the “Stop **<u>Sign</u>**” on JMRI throttle and then close and reopen DecoderPro to start again.

To release the loco, hit the “Release: button on the JMRI throttle.

The first Speed step table bar is associated with CV67 and Speed Step #1 in the 0-28 DCC speed step scale. Last Speed Step table bar is associated with CV94 and Speed Step #28 in the 0-28 DCC speed step scale.

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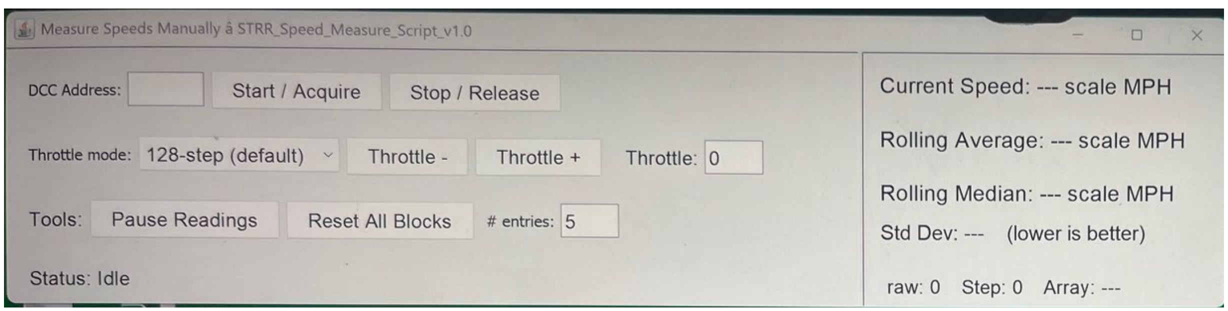
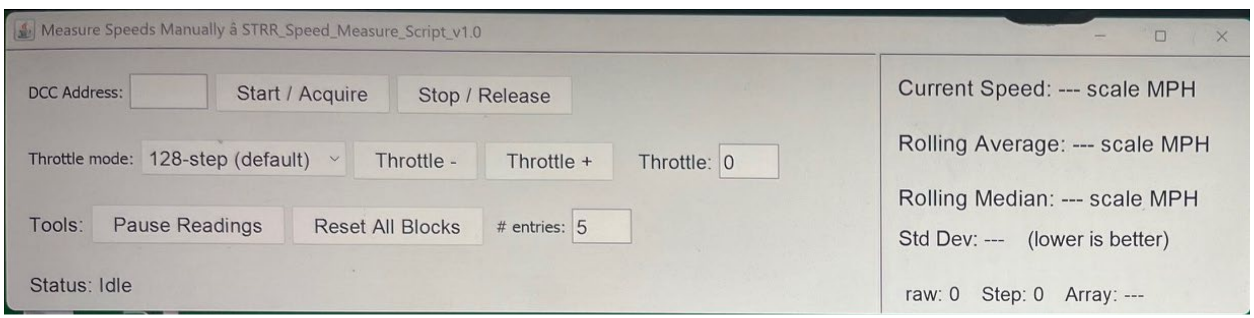
Step 8	There are 3 options to control a locomotive on the loop of track when using this Speed Measuring script. **<u>Option #1</u>** - You can use the <u>virtual JMRI throttle</u> as described in Step 6 above. You can select from the drop-down menu on the throttle a scale of 0-28 or 0-128. The scale 0-128 gives you a wider range of control. **<u>Option #2</u>** – You can use a <u>physical Digitrax throttle</u> (such as DT204) plugged into the LocoNet cable port "B" on the DCS 100 box. You need to acquire the loco on the throttle to control it. **<u>Option #3</u>** – You type the DCC 4-digit loco address in the <u>speed measuring input window</u> in the "**DCC Address**" text box and select "**Start/Acquire**" button. You can control the speed of the loco via the "**+/- Throttle**" buttons. 	You can view the Digitrax throttle % vs. 0-128 scale vs. 0-28 scale in the <u>Appendix #1</u> at end of this document. Using the 0-128 scale can provide you with more resolution between throttle percentages.

Step 9	The following buttons on the Speed Measuring script input do the following actions: DCC Address – Where you input your 4-digit DCC Locomotive address. Start / Acquire – Button creates a virtual throttle with your DCC Loco address. Stop / Release – Button stops the loco and releases the address from the DCC system. Throttle Mode – Change between using speed step 0-128, 0-28, or percentage from 0-99%. +/- Throttle – Controls the speed of the loco Throttle – Current throttle setting based on Throttle Mode selected. Pause Readings – Stops script from taking readings temporality. Reset All Blocks – Resets the values measured in the Speed Measurement window. # entries – The number of entries used for the Rolling Average, Rolling Median, and Standard Deviation values. Default is 5 values. Raw # - Throttle value between 0-127. Step # - Throttle value between 0-28. Array - # of blocks used to capture a measurement. 	For Arrays: MaxSpeedArray = 4 blocks used for measurement HighSpeedArray = 3 blocks used for measurement MediumSpeedArray = 2 blocks used for measurements LowSpeedArray = every block used for measurements

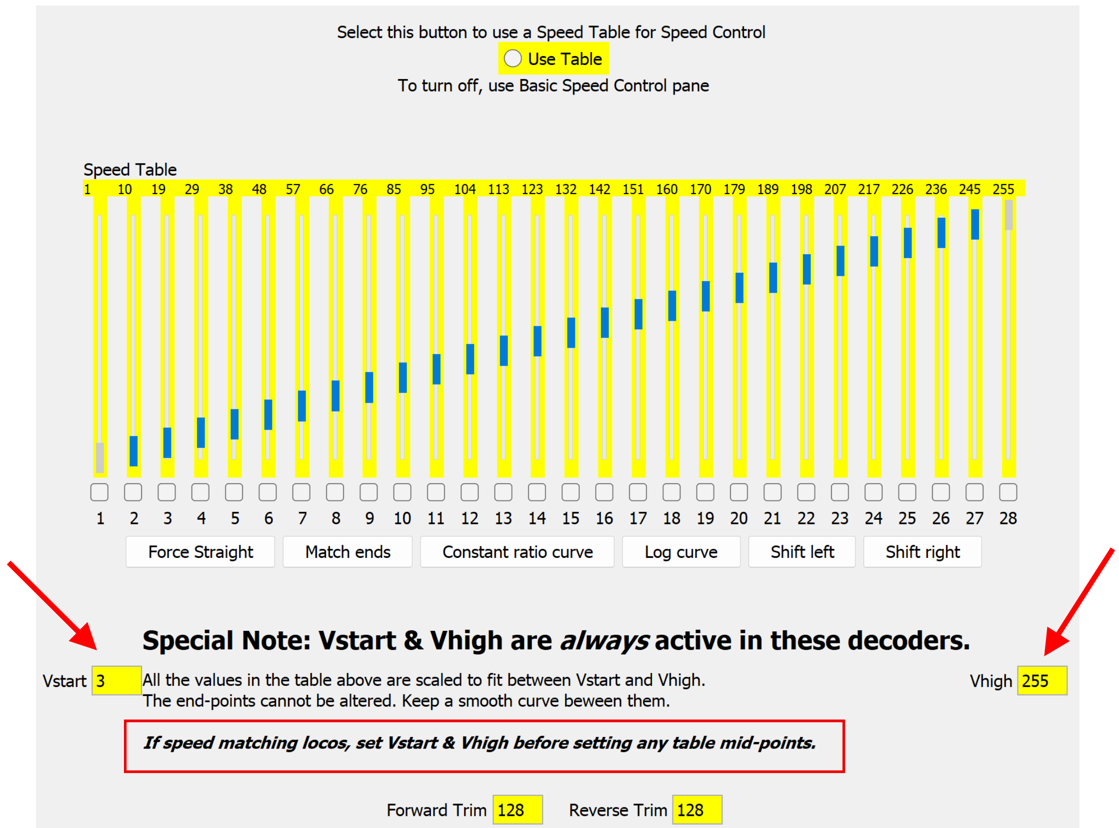
Step 10

Digitrax Throttle %	128 Speed Steps	28 Speed Steps	Speed Table CV Settings	Target Speed Values
0	0	0	--	0
1	1	1	67	Initial Movement
4	5	2	68	4
8	10	3	69	8
11	14	4	70	11
14	18	5	71	14
18	23	6	72	18
22	28	7	73	22
26	33	8	74	26
30	38	9	75	30
33	42	10	76	33
36	46	11	77	36
40	51	12	78	40
44	56	13	79	44
48	62	14	80	48
52	67	15	81	52
55	71	16	82	55
58	74	17	83	58
62	80	18	84	62
66	85	19	85	66
70	90	20	86	70
74	95	21	87	74
77	99	22	88	77
80	103	23	89	80
84	108	24	90	84
88	113	25	91	88
92	118	26	92	92
96	123	27	93	96
99	127	28	94	99

The Gray boxes are the throttle measurements that are taken for the **Speed Matching** Script.

Step 11	To manually program your DCC locomotive, you will want to take the following steps: A) Make sure the key windows on laptop are opened and arranged as shown in Step #7. Best to have **POM** radial dial selected to make DCC decoder chip changes on the loop of track when using DecoderPro. B) Decide which throttle option you are going to use and program the throttle for the DCC loco you wish to measure the speed on the loop of track. C) Place loco on loop of track so it moves in a clockwise direction. D) Using the table in Step #10, select a throttle percentage or speed step to program. Note the 0-28 speed step value of the speed value you are trying to measure and program. E) Select the throttle percentage or speed step using the throttle option you selected. You will see "Raw" or "Speed Step" value change to the value you desired in the Speed Measurement Script window. F) Hit the "**Reset All Blocks**" button to clear out values in script window and get fresh measurements. The longer the program runs, the better the measurements will stabilize. G) Observe the speed values being measured on the Speed Measurement Script window. Use the Rolling Average or Rolling Median along with the Standard Deviation to determine if you need to make further adjustments. H) Using DecoderPro Speed Table tab that should be already opened, locate the appropriate 0-28 speed step bar and adjust the bar up/down and then select "**Write Changes on Sheet**" button. (This can be done while the loco is moving!) I) Repeat Steps #E thru #G for the rest of the desired throttle values you wish to program. You may be able to estimate what the values should be between 2 measured values. J) Use your throttle to confirm measured speeds (and those estimated in between) align with throttle percentages or speed steps.	Using 0-128 speed step scale will be best for this manual programming step because of the resolution it offers compared to the 0-28 speed step scale. Every speed step of 0-28 scale equals 4.7 units on the 0-128 speed step scale. You can see this if you select 0-28 speed steps on JMRI virtual throttle and then view physical Digitrax throttle % changes.

<table>
<tr><td>Step 12</td><td>Clicking on the “<u>Stop Sign</u>” on JMRI throttle or the “Stop / Release” button on Speed Measurement script window will cause the loco to stop. You may need to close and reopen DecoderPro to start again.</td><td></td></tr>
<tr><td>Step 13</td><td>When speed measuring a DCC locomotive with a SOUND Decoder, you may need to set CV67 (initial movement) and CV94 (Max speed) <u>FIRST</u> before evaluating the other speed steps in the table. <u>Non-Sound decoders usually do not have this issue.</u> If your Speed Table in DecoderPro looks like the one below, you will need to do this.

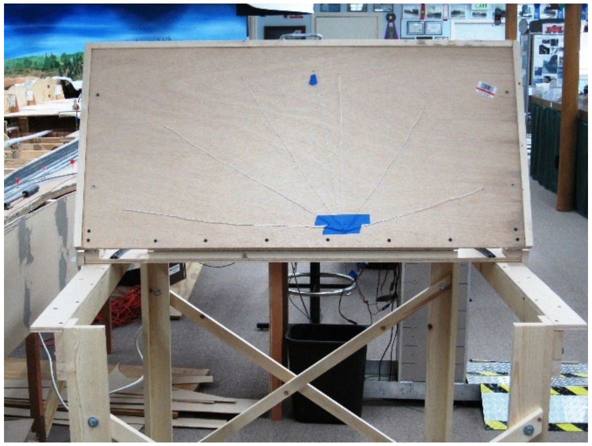

</td><td>Many LokSound sound decoders and some Tsunami sound decoders have this speed table feature. Others do not. The best way to know this is to select the appropriate DCC manufacturer and decoder type in DecoderPro when opening up a “New Loco”.</td></tr>
</table>

Step 14	In DecoderPro, in the Comprehensive Program, under the tab "**CVs**", one can see all the values recorded for each CV. By selecting "**Read full sheet**", you will see each CV value recorded in the DCC Decoder. When **CV29** is set to "**50**", it means the following: → 28/128 Speed Table Enabled → DC Analog Disabled ➔ Normal direction of travel is Forward. → 4-Digit DCC Address being used. →*Reverse direction* sets CV 29 to **51**. ***SEE STEP 15 FOR OTHER VALUES FOR CV29 AND THEIR EFFECTS.***	

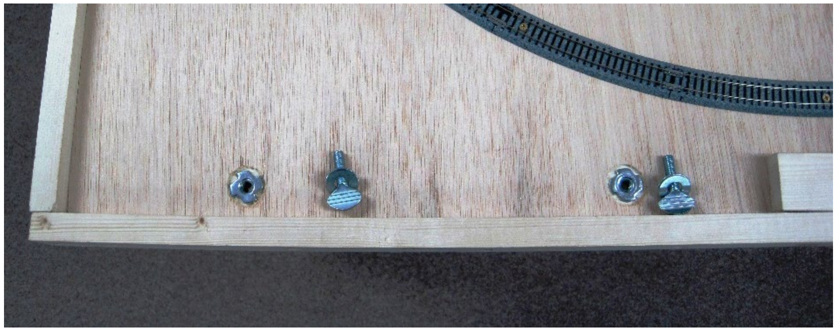
Step 15

CV Value for CV29	Speed Steps / Speed Table	Analog Mode Conversion	Normal Direction of Travel	2- or 4-Digit Address
006	28/128	ON	Forward	2
007	28/128	ON	Reverse	2
016	14 Speed Table	OFF	Forward	2
017	14 Speed Table	OFF	Reverse	2
018	28/128 Speed Table	OFF	Forward	2
019	28/128 Speed Table	OFF	Reverse	2
020	14 Speed Table	ON	Forward	2
021	14 Speed Table	ON	Reverse	2
022	28/128 Speed Table	ON	Forward	2
023	28/128 Speed Table	ON	Reverse	2
032	14	OFF	Forward	4
033	14	OFF	Reverse	4
034	28/128	OFF	Forward	4
035	28/128	OFF	Reverse	4
036	14	ON	Forward	4
037	14	ON	Reverse	4
038	28/128	ON	Forward	4
039	28/128	ON	Reverse	4
048	14 Speed Table	OFF	Forward	4
049	14 Speed Table	OFF	Reverse	4
050	28/128 Speed Table	OFF	Forward	4
051	28/128 Speed Table	OFF	Reverse	4
052	14 Speed Table	ON	Forward	4
053	14 Speed Table	ON	Reverse	4
054	28/128 Speed Table	ON	Forward	4
055	28/128 Speed Table	ON	Reverse	4

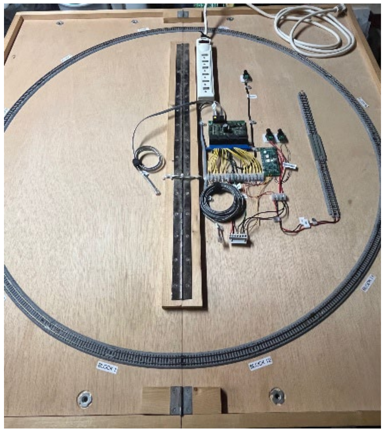
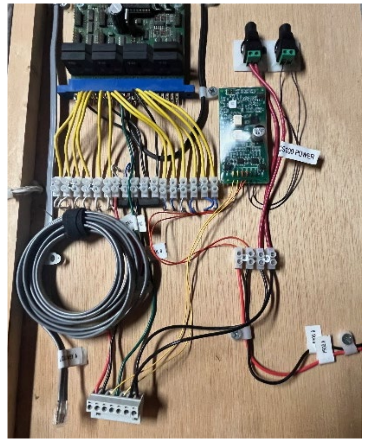
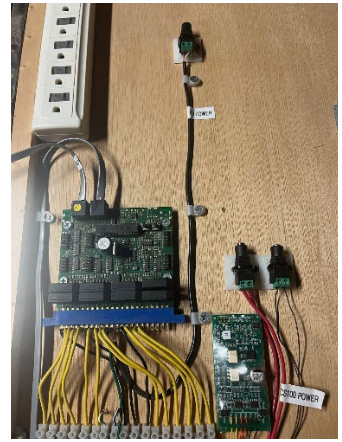
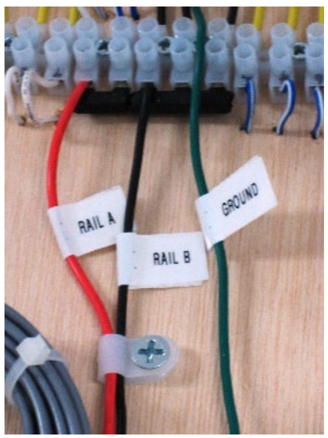
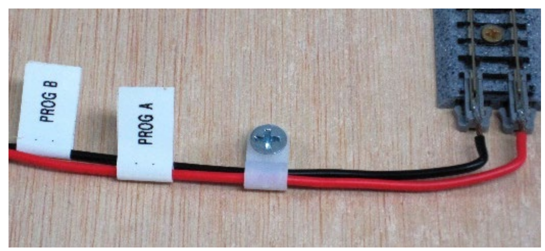
<u>Section 3:</u> Assembly of the STRR Speed Matching Table (SMT)
(Skip Steps 1 & 2 if using STRR SMT without the support stand.)

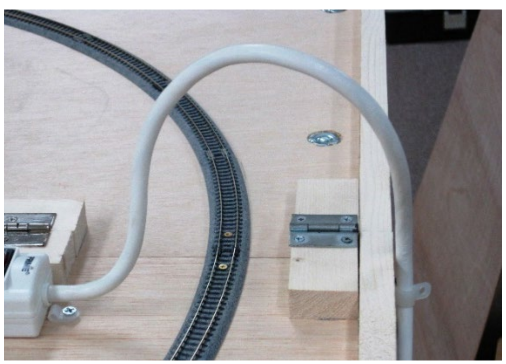
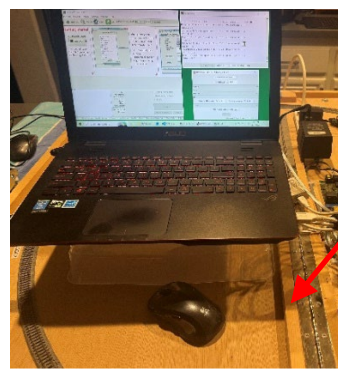
Steps	STRR SMT Assembly Directions	Notes
Step 1	Place the SMT on top of the L-girder benchwork such that the metal handle is towards one of the ends with the "**X**" cross beams. Arrange it such that the thumb screws of the handle are on top.   	

<table>
<tr>
<td>Step 2</td>
<td>Undo the thumb screws on the handle and put them back in after lifting up the table a few inches. Unfold the table all the way and attach SMT to the L-girder benchwork using 8 thumbscrews and washers from underneath.

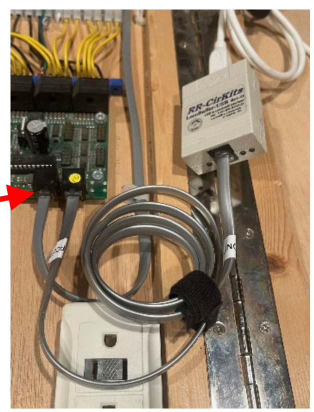


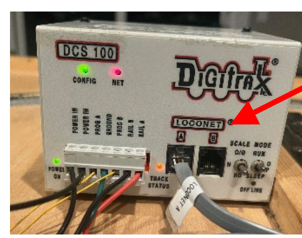
</td>
<td>If the 8 holes do not align properly, fold SMT back up and rotate 180 degrees so the handle is at the other end of the benchwork with the “X” cross beams.</td>
</tr>
</table>

Step 3	Arrange the SMT such that the surge protector is towards the back. Familiarize yourself with the location of the different wires/cords on the right side of the table.       	You will need to know the location of the LocoNet A cable, LocoBuffer-USB Cable, Rail A & Rail B wires, Ground, Prog A & Prog B wires, and the power cord connectors for the Programming Track Booster (PTB) circuit board, DCS 100, and the Block Detection (BD) circuit board.

Step 4	Locate and familiarize yourself with the following items:     	
Step 5	Take the surge protector power cord and bend it such that it creates an arch over the track. Place the cord into the plastic clip in the back and plug it into a nearby outlet. ***<u>MAKE SURE SURGE PROTECTOR IS TURNED OFF!</u>*** Take the laptop, laptop power cord, mouse, and the LocoBuffer-USB device with the gray cable and place them on the ***<u>LEFT-SIDE</u>*** of the SMT hinge. The laptop can sit on top of the bin that stores the throttle and cords. You can plug the mouse and LocoBuffer-USB device into any of the USB ports on the laptop. Plug the laptop power cord into both the laptop and the surge protector on the SMT. ***<u>See pictures below for details</u>***.  	The LocoBuffer-USB device can sit on the left side of the hinge next to the laptop. Plugging the LocoBuffer-USB device into the USB port on the <u>right side of laptop</u> is best.

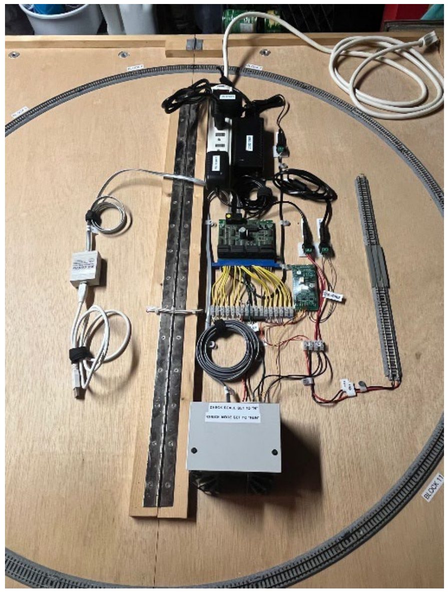
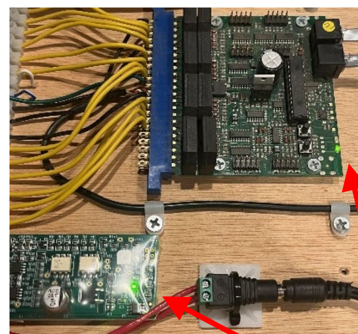
<table>
<tr>
<td>Step 6</td>
<td>Place the Digitrax DCS_100 in front of the grey connector containing the Rail A & B, Ground, and other wires. Plug this grey multi-wire connector into the DCS_100, making sure that the wires match up with the labels on the DCS_100. Plug the grey <u>LocoNet A cable</u> into the <u>LocoNet A port</u> on the back of the DCS_100. <u>See pictures below for details</u>.


</td>
<td></td>
</tr>
<tr>
<td>Step 7</td>
<td>Plug the gray LocoNet USB-LocoBuffer cable into <u>EITHER</u> the Rail B port of the DCS_100 <u>OR</u> into the 2nd port of the BDL168 block detection board (preferred).   <u>See pictures below for details</u>.
BDL168 Block Detection LocoNet Port (preferred)




LocoNet Rail B Port on back of DCS_100
</td>
<td>If you wish, you can plug another Digitrax throttle (i.e. DT400R) into the LocoNet B port to use for programming and testing later.</td>
</tr>
</table>

Step 8	Plug the Digitrax DCS_100 power cord into <u>the matching connector</u> on the SMT table. Plug the BDL168 Block Detection (BD) circuit board power cord connector into <u>the matching connector</u> on the SMT table. Plug the PTB-100 Programming Track Booster circuit board power cord connector into <u>the matching connector</u> on the SMT table.    	

Step 9	The assembly of the STRR SMT is now complete and ready to be used to speed match or speed measure DCC locomotives. Turn on the power button on the surge protector to begin using the table for DCC locomotives. ***<u>See Section 1 for using the STRR Speed Table Software to Speed Match DCC Locomotives.</u>***  	Turn on the power strip to turn everything on. The power blocks in the power strip will have lights on them. The BDL168 circuit board itself will have 2 lights (red & green) turn on and flash a few times. The DCS_100 will have a green light that flashes 1x per 4 seconds.

Appendix #1

Digitrax Throttle %	128 Speed Steps	28 Speed Steps	Speed Table CV Settings	Target Speed Values
0	0	0	--	0
1	1	1	67	Initial Movement
2	3	2		
3	4			
4	5		68	4
5	6			
6	8	3		
7	9			
8	10		69	8
9	12			
10	13	4		
11	14		70	11
12	15			
13	17	5		
14	18		71	14
15	19			
16	21			
17	22	6		
18	23		72	18
19	24			
20	26			
21	27	7		
22	28		73	22
23	30			
24	31	8		
25	32			
26	33		74	26
27	35			

The table shows the differences of throttle % (Orange) vs. 128 Speed Steps (Ylw) vs. 28 Speed Steps (Green) vs. Speed Table CV settings (Blue) vs. Target Speed Values (Pink).

Using 0-28 speed step scale will allow CV67–94 values to map directly one-to-one with unique throttle positions.

		28	36	9			Using 0-128 speed step scale will allow CV67–94 values to still define the curve shape, but the decoder will interpolate between the CV values to produce smoother sub steps for the different throttle positions.
		29	37				
		30	38		75	30	
		31	40				
		32	41	10			
		33	42		76	33	
		34	44				
		35	45	11			
		36	46		77	36	
		37	47				
		38	49				
		39	50	12			
		40	51		78	40	
		41	53				
		42	54				
		43	55	13			
		44	56		79	44	
		45	58				
		46	59	14			
		47	60				
		48	62		80	48	
		49	63				
		50	64	15			
		51	65				
		52	67		81	52	
		53	68				
		54	69	16			
		55	71		82	55	
		56	72				
		57	73	17			
		58	74		83	58	
		59	76				
		60	77				

61	78			
62	80	18	84	62
63	81			
64	82			
65	83			
66	85	19	85	66
67	86			
68	87			
69	89			
70	90	20	86	70
71	91			
72	92			
73	94			
74	95	21	87	74
75	96			
76	97			
77	99	22	88	77
78	100			
79	101			
80	103	23	89	80
81	104			
82	105			
83	106			
84	108	24	90	84
85	109			
86	110			
87	112			
88	113	25	91	88
89	114			
90	115			
91	117			
92	118	26	92	92
93	119			

94	121	27		
95	122			
96	123		93	96
97	124			
98	126	28		
99	127		94	99